

Jashan Singh

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TECHNICAL SKILLS

Languages: C++, Python, Java, JavaScript, TypeScript, SQL (PostgreSQL)

Backend & Systems: Node.js, Express.js, FastAPI, Flask, POSIX Sockets, Multithreading, TCP/IP, WebSockets, REST APIs

Frontend: React, Vue.js, Tailwind CSS, MUI

Data, Cloud & Security: AWS, Pandas, NumPy, Jenkins, libsodium

Tools & Collaboration: Git, GitHub, Jira

EDUCATION

Bachelor of Software Engineering

Sep 2025 – Present | Waterloo, ON

University of Waterloo

Systems & Programming: Programming Principles (C), Data Abstraction (C++), Digital Circuits (Verilog), Linear Circuits

Mathematics: Linear Algebra, Calculus I & II, Multivariate Calculus, Discrete Mathematics

EXPERIENCE

Lount {Rental Property Management Tool}

Apr 2026 – Present | Remote

Software Engineer

- Developed and deployed a full-stack Next.js/TypeScript SaaS platform to automate rent collection and maintenance ticketing for regional property investors.
- Architected dynamic backend billing logic using the Stripe API and Supabase PostgreSQL triggers to automatically synchronize multi-tenant subscription states and compute prorated cycles based on active unit capacities.
- Engineered a server-side document generation pipeline that dynamically compiles real-time maintenance logs and rent ledgers into Bill 60-compliant evidentiary PDFs for Landlord and Tenant Board (LTB) disputes.

Rinknet Software {Hockey Scouting}

May 2025 – Sep 2025 | Waterloo, ON

Software Engineering Intern

- Engineered scalable Python data pipelines and Selenium scrapers to autonomously extract complex JavaScript-rendered data, reducing manual processing time by 90%
- Managed configuration and deployment of CI/CD workflows using Jenkins and Windows Task Scheduler, ensuring high availability and 99% uptime for production-critical data ingestion
- Validated system performance through rigorous software testing and automated alerting, providing real-time reporting on pipeline health
- Collaborated within an Agile framework to translate scouting requirements into functional technical designs, accelerating the development lifecycle for new data modules

Sera4 {Keyless Access Management}

Sep 2024 – Feb 2025 | Waterloo, ON

Software Engineering Intern

- Architected a scalable front-end system by creating reusable MUI components and structuring CSS architecture for maintainable UI development and reduced frontend production time by 40%
- Built an interactive data dashboard, efficiently fetching, sorting, and visualizing large datasets using React and graphical libraries
- Collaborated in an Agile/Scrum environment using GitHub for version control and Jira for sprint planning, contributing to streamlined team-based development and project delivery

PROJECTS

ZenithDrop

C++, TCP, Sockets, Libsodium, Diffie-Hellman, HTTP, Linux / Unix

- Architected a concurrent TCP server utilizing POSIX sockets and a custom Thread Pool; implemented `std::condition_variable` and `std::unique_lock` to efficiently synchronize client streams and eliminate CPU busy-waiting.
- Developed a robust length-prefix framing protocol to resolve TCP stream fragmentation, ensuring precise payload alignment for authenticated encryption.
- Secured network payloads using libsodium XChaCha20-Poly1305 cryptography; achieved $O(1)$ dynamic memory allocation by processing data in fixed 1KB blocks streamed directly to disk.
- Integrated a zero-configuration HTTP server and a terminal-native QR code generator using POSIX `ifaddrs` to facilitate seamless cross-device pairing and localized file distribution.

Event Driven Algorithmic Trading Engine

Python, Tiingo API, Interactive Brokers, FinBERT

- Engineered an autonomous, event-driven trading architecture to ingest real-time market data and evaluate dynamic signal conditions utilizing FinBERT sentiment analysis.
- Architected a robust order-execution state machine featuring a dynamic trailing stop-loss algorithm to calculate and mitigate high-frequency execution slippage.
- Programmed strict time-decay evaluation constraints to automatically compress risk parameters and force position liquidation prior to end-of-day liquidity drops.

Technical Systems Architecture Series: codingdemon

C++, Linux/Unix System Programming, Technical Documentation

- Producing technical educational content focusing on Linux/Unix System Programming, applying the Feynman Technique to document complex architectures.
- Breaking down low-level concepts including process control, memory management, and signal handlers into accessible developer resources.